Alias: Corbato

Homework 2 Functions

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Unset
Main:   add  $t0,$s1,$s1      # $t0 ← 2y
        add  $t0,$t0,$t0      # $t0 ← 4y
        addi $t0,$t0,4        # $t0 ← 4y+4
        addi $sp,$sp,-4       # Push $t0
        sw   $t0,0($sp)
        add  $a0,$t0,$zero    # arg0 ← 4y+4
        jal  Foo              # Call foo
        lw   $t0,0($sp )      # Pop $t0
        addi $sp,$sp,4        #
        add  $s0,$v0,$v0      # x ← 2*foo(4*y+4)
        add  $s0,$t0,$s0      # x ← x+4*y+4
        ...                   # End of Main
Foo:    addi $sp,$sp,-104     # (MOD 1) Stack space for $ra, $fp,
                              # magic, junk
        sw   $ra,100($sp)     # Push $ra
        sw   $fp,96($sp)      # Push $fp
        addi $fp,$sp,100      # Establish frame
        addi $t1,$zero,0      # $t1 ← 0
        sw   $t1,-44($fp)     # junk[0] ← 0
        addi $t1,$zero,1      # $t1 ← 1
        sw   $t1,-40($fp)     # junk[1] ← 1
        ...
        addi $t1,$zero,9      # $t1 ← 9
        sw   $t1,-8($fp)      # junk[9] ← 9

        lw   $t1,-28($fp)     # $t1 ← junk[4]
        lw   $t2,-68($fp)     # (MOD 2) $t2 ← magic[8]
        sub  $t2,$zero,$t2    # $t2 ← -magic[8]
        addi $t2,$t2,2        # $t2 ← 2-magic[8]
        add  $t1,$t1,$t2      # $t1 ← junk[4]+2-magic[8]
        sw   $t1,-28($fp)     # junk[4] ← junk[4]+2-magic[8]
        slti $t0,$a0,1        # $t0 ← 1 if n<1; else 0
        bne  $t0,$zero,RetOne # Return 1 if n<1
```

```
# (MOD 3) Call bar(n+1,magic[8],n+2,n+3,magic[8])
        addi $sp,$sp,-20      # Stack space for 5 args
        addi $t0,$a0,1        # $t0 ← n+1
        sw   $t0,0($sp)       # Push n+1
        lw   $t1,-68($fp)     # $t1 ← magic[8]
        sw   $t1,4($sp)       # Push magic[8]
        addi $t0,$t0,1        # $t0 ← n+2
        sw   $t0,8($sp)       # Push n+2
        addi $t0,$t0,1        # $t0 ← n+3
        sw   $t0,12($sp)      # Push n+3
        sw   $t1,16($sp)      # Push magic[8]
        jal  Bar              # Call bar(n+1,magic[8],n+2,n+3,magic[8])
        add  $t0,$v0,$v0      # $t0 ←
                              # 2*bar(n+1,magic[8],n+2,n+3,magic[8])
        addi $sp,$sp,20       # reclaim stack space
# Call foo(n-1)
        addi $sp,$sp,-4       # Push
        sw   $a0,0($sp)       # ... $a0
        addi $a0,$a0,-1       # arg0 ← n-1
        jal  Foo              # Call foo(n-1)
        lw   $a0,0($sp)       # Pop $a0
        addi $sp,$sp,4
        lw   $t1,-28($fp)     # $t1 ← junk[4]
        add  $v0,$v0,$t1      # $v0 ← foo(n-1)+junk[4]
        add  $v0,$v0,$t0      # (MOD 3) $v0 ← foo(n-1)+junk[4]+
                              # 2*bar(n+1,magic[8],n+2,n+3,magic[8])
        j    Ret              # Return
RetOne: addi $v0,$zero,1      # $v0 ← 1
Ret:    lw   $fp,96($sp)      # (MOD 1) pop $fp
        lw   $ra,100($sp)     # pop $ra
        addi $sp,$sp,104      # reclaim stack space
        jr   $ra              # return to caller
```

Did not use $a0-$a3 variables for bar